IAIN CHESWORTH LABS · HUMAN FACTORS & DESIGN

Word Jewels Solver.

A deep human-factors review of the running application — and the design overhaul it needs to feel like a finished product, not a prototype.

19 October 2026 Windows · Qt Quick 6.12 · 1116 × 739 42 QML files · 6,777 lines Read the full audit · PDF

7
critical or high findings in the core child flow

1.3
/4
heuristic score: prototype-level, not release-level

0
visible labels on the home screen's two biggest choices

1.06:1
contrast of the word path's casing on a die

THE VERDICT

The engine is ready.
The product surface is the work.

The game underneath is real: a fast solver, a thoughtful privacy model, generous targets and a genuinely good scan-and-confirm flow. But the first screen is wordless, the board is a 3D scene pasted next to empty cards, the jewels are almost invisible, and the parent area is a second product. This is not a palette problem. It needs a design system, a clear split between what a child does and what a grown-up does, and a feedback loop that makes a five-year-old feel clever.

3
visual vocabularies: child kit, parent kit, 3D scene

2
button systems: square outline vs rounded fill

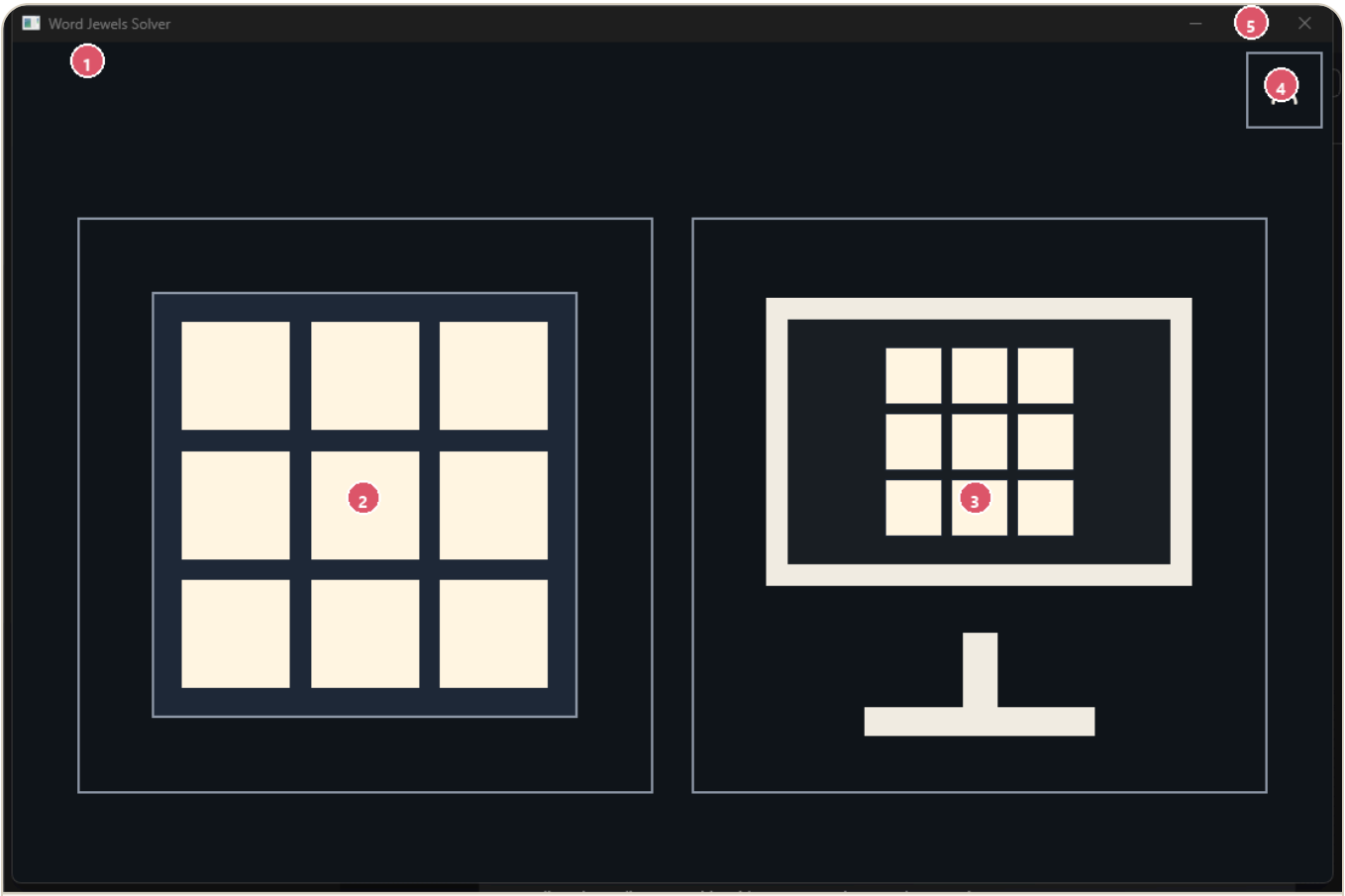
0
app icons, wordmarks or bundled learning fonts

3 s
after the last word, the app moves on without a summary

EVIDENCE · THE FIRST THIRTY SECONDS

A child sees two pictures. A parent sees a puzzle.

The home screen is the product's front door. It has no wordmark, no app name, no tagline and no visible mode labels. The two largest choices are a grid and a monitor.



Home: the first thirty seconds

1. No wordmark, app name or tagline: the only name is the OS title bar.
2. A wordless picture of a grid: is it play, solve or continue?
3. A second wordless picture that reads as a monitor, not a family game.
4. The only way to the parent area is an unlabelled person icon.
5. Generic document icon and system title bar; no product identity.

Exhibit 1. Home, as it runs. The numbers are explained under the image in the full audit. The only name is the OS title bar; the only route to the parent area is an unlabelled person icon.

A FIVE-YEAR-OLD

“Which one is play?”

Two wordless cards. One looks like a grid, one like a monitor. No play triangle, no hand, no spoken label until after the choice. The child has to guess, or ask an adult. The adult has no visible instructions either.

A PARENT

“Where are the settings?”

The family mode looks like a monitor, not a camera or a real board. Choosing it leads to a no-camera screen and a 16-cell typing form. The dashboard, sound and language are behind an unlabelled person icon and a two-circle gate.

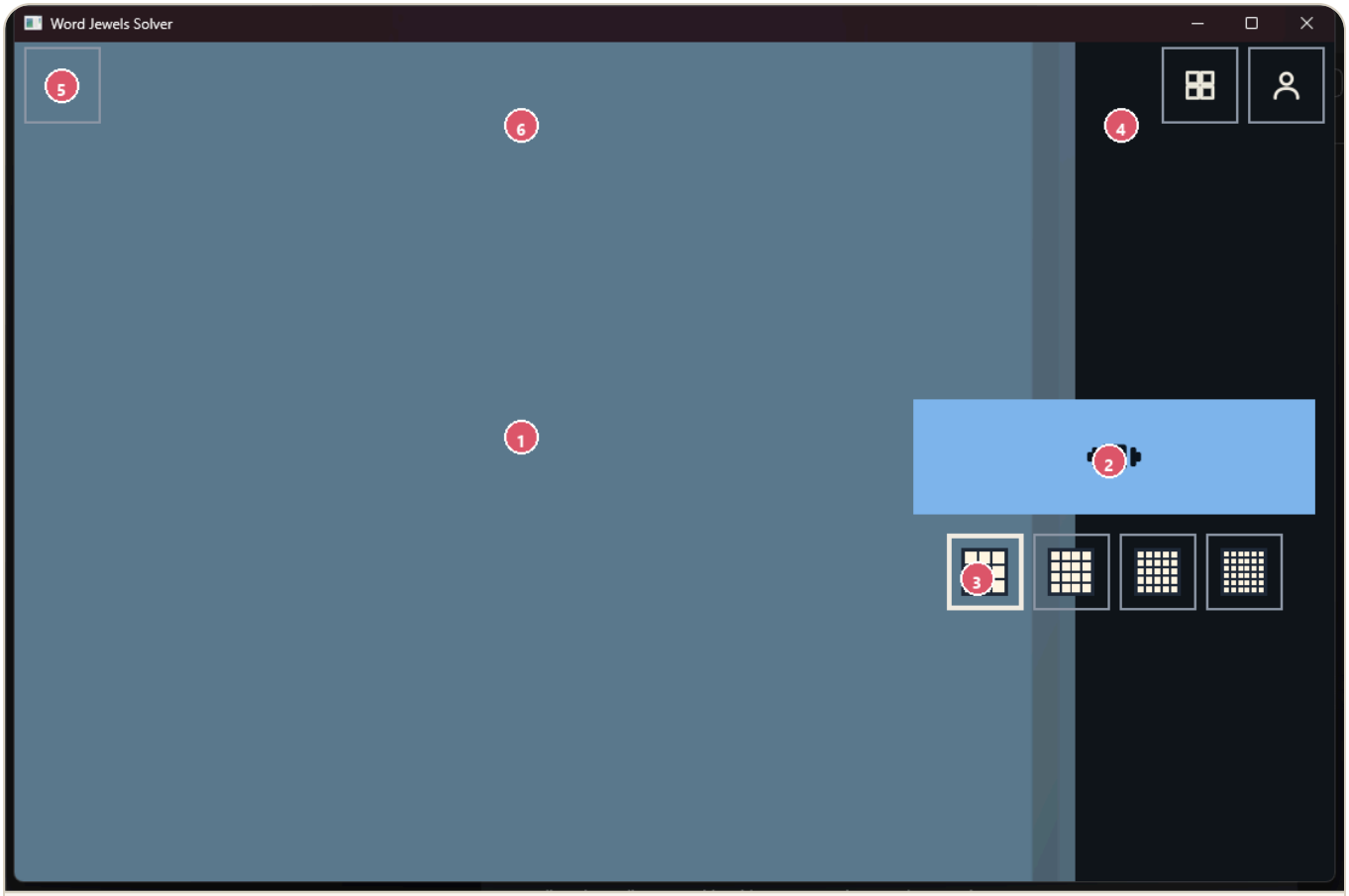
The role problem. Solo Play is a child's game. The Family Companion is an adult-led setup: place a board, capture it, check 16 letters, confirm. The home treats them as two equal child choices and hides the adult's settings. That is upside down.



Solo playing: a board pasted next to empty boxes

1. The 3D board deck is slate blue in both themes; the cream chrome ignores it.
2. Heavy black seams, uneven perspective and drop shadows fight the letters.
3. The largest panel card has no title, count or empty state; it looks broken.
4. A second blank card, with an icon that could mean grid, apps or history.
5. A large amber card pulls the eye away from the board for a secondary action.
6. The hint is a tiny ring-and-dot; nothing says 'show me a start'.
7. No status, score, jewel count or sense of progress anywhere on the screen.

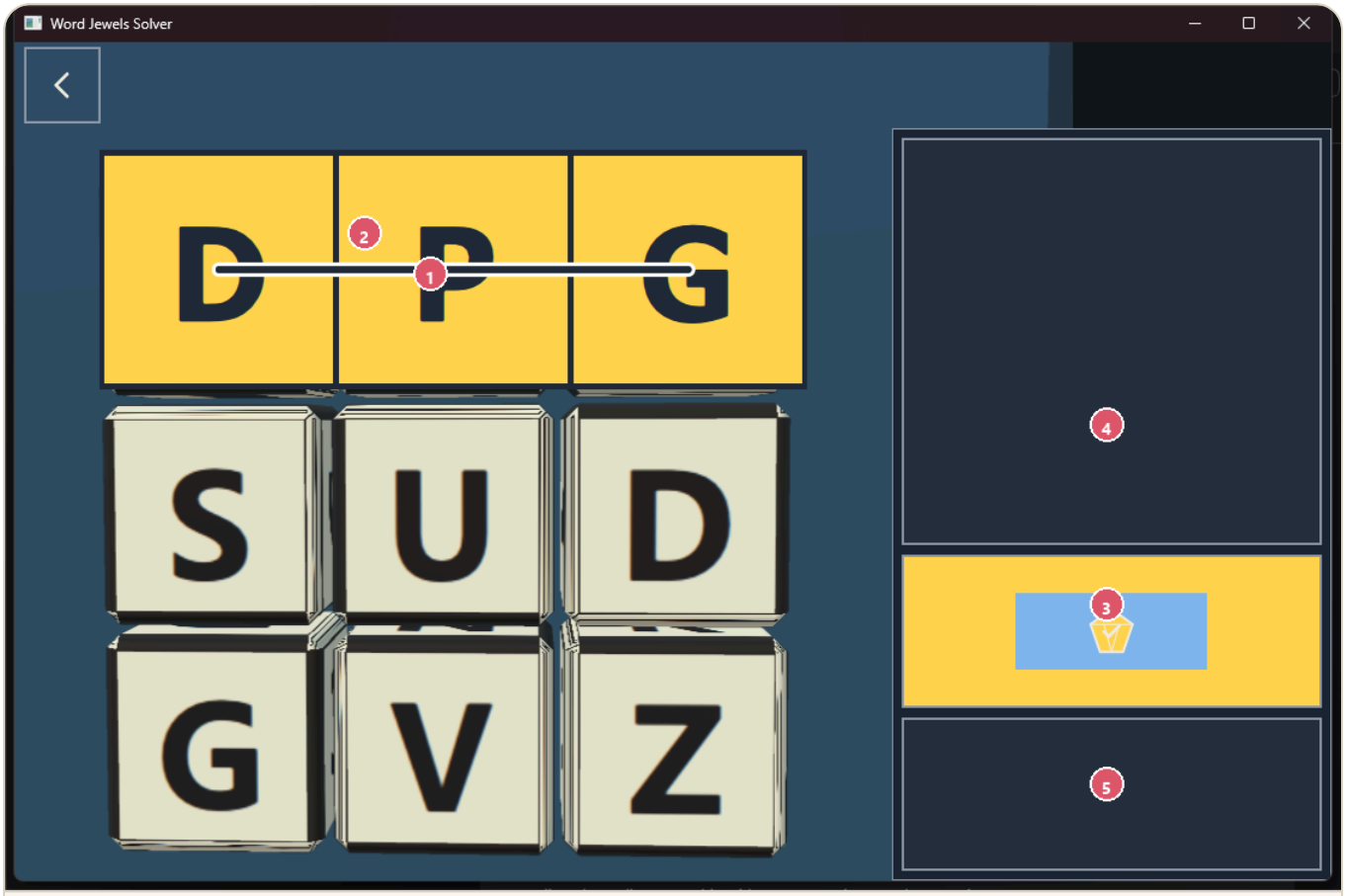
Exhibit 2. Solo playing. The board is readable, but it belongs to a different product from the cream chrome and the blank panel. The largest card has no title or empty state; the amber hint card pulls the eye away from the board.



Solo box: the core metaphor is missing

1. The box with its lid is invisible: a flat slate field is all that renders.
2. One huge unlabelled Shake button; the icon reads as a speaker or battery.
3. Four near-identical grid icons for 3x3 to 6x6; no text or size cue.
4. The panel's place is a blank white rectangle with no title or purpose.
5. Back floats over the play area rather than living in a stable app bar.
6. No title, no instruction, no indication of what happens next.

Exhibit 3. Solo box. The box with its lid — the core metaphor — is a flat slate field on the review machine. The Shake control is a banner with a speaker-like icon; the four size grids are indistinguishable at 64 px.



Selecting a word: the interaction works, the communication does not

1. The path line is drawn through the letters, cutting D, P and G in half.
2. A lit die becomes a flat yellow rectangle: the die, edge and shadow vanish.
3. The only way to submit is an icon-only Done button that replaces the hint.
4. The word being built is not shown; the child has to remember the path.
5. There is no visible Clear/Cancel; a mistake means starting again by trial.

Exhibit 4. Selecting a word. The path cuts through D, P and G. Lit dice become flat yellow rectangles. The only way to submit is an icon-only gem that replaces the hint.

THE FINDINGS

Seven things a design director would stop the line for.

Each finding is a product problem, not a preference. The evidence and the code cause are in the full audit; the fix is in the direction below.

01 / CRITICAL

CRITICAL

No identity, no design system.

No wordmark, no app icon, no bundled typeface, no single theme, no shared app bar. The title bar shows a generic document icon; the child kit is square and outline-only; the parent kit is rounded and filled; the 3D scene has its own colours.

Fix: one wordmark, one icon, one type system, one palette, one component kit. The parent area is the same product, denser.

CRITICAL

02

The home is a wordless, mis-targeted mode picker.

Two huge unlabelled pictures ask a pre-reader to choose between a game and an adult scanning workflow. The good names — “Play on my own”, “Play with the family” — exist only for the screen reader.

Fix: label the cards, make Play primary, label For grown-ups, and put the adult workflow behind the gate.

CRITICAL

03

The board and the panel are disconnected.

The box is invisible on the review build. The board sits on a slate 3D deck that ignores the theme. The panel is three blank cards. The 3D seams and perspective fight the letters.

Fix: a framed board stage; a flat, crisp play surface; titled panel cards with empty states.

HIGH

04

The loop has no visible reward or ending.

A found word adds a small list row. Clearing the board produces a glow, then a three-second move to a hidden hold-to-play screen. There is no “you found all the words”, no jewel summary, no praise.

Fix: a jewel for every word, a spoken word, a clear summary, visible Play again and Home.

HIGH

05

Child controls are icon-only and often silent.

Four grids, a ring-and-dot, a person, a gem with a tick. Only three of 29 buttons set the spoken label. A five-year-old cannot decode the interface; a parent cannot help.

Fix: a visible word and a spoken word on every child control; test the icons with children.

HIGH

06

The parent area is a second product.

Rounded buttons, 12 px radii and a white bar in the parent area; square outlines in the child area. Settings are one unbroken list ordered by implementation. The dashboard empty state is a dead end.

Fix: one design system; five settings groups; a welcome card; privacy at the top; a compact gate.

HIGH

07

The app is not productised.

Fixed 1100 × 700 window, top-left, no persistence, no loading state, no icon, Andika named but not bundled, no line between the OS frame and the app.

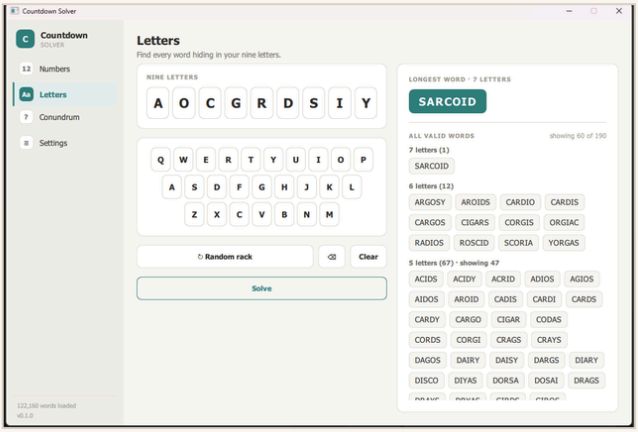
Fix: centre and remember the window; branded first paint; bundle the fonts; a real app icon; a 2D board fallback.

THE HOUSE COMPARISON

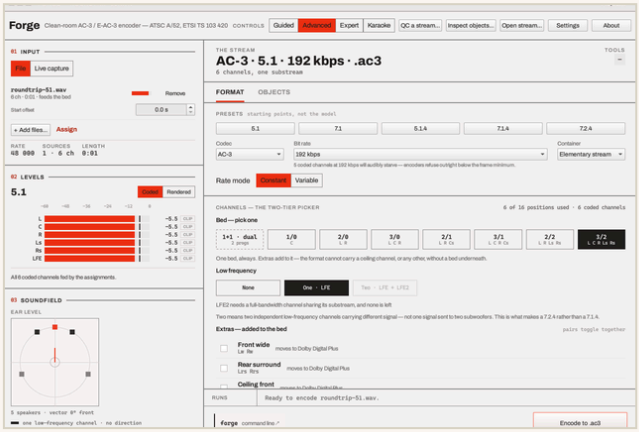
Same house. Different discipline.

Countdown Solver and ICL Forge do not look alike, but they share a discipline: a wordmark, a type system, one accent, labelled controls, clear hierarchy and real empty states. WordJewels has the opposite profile.

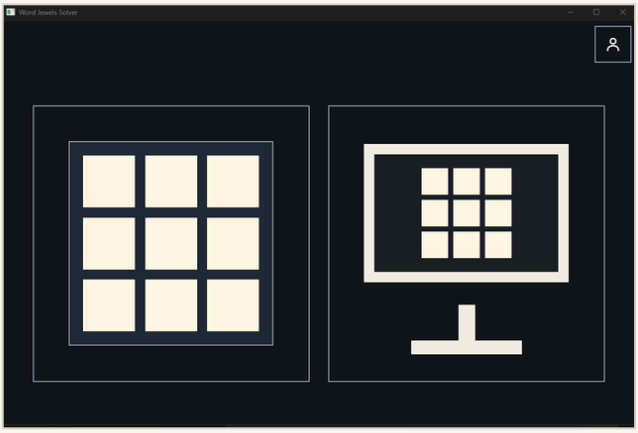
Countdown Solver
Wordmark, sidebar, clear page title, labelled controls, one teal accent.



ICL Forge
Wordmark, numbered sections, mono labels, one red accent, clear hierarchy.



WordJewelsSolver: home
No wordmark or labels; two wordless cards; one tiny parent icon.



WordJewelsSolver: Solo Play
Board pasted next to blank cards; icon-only controls; no progress.

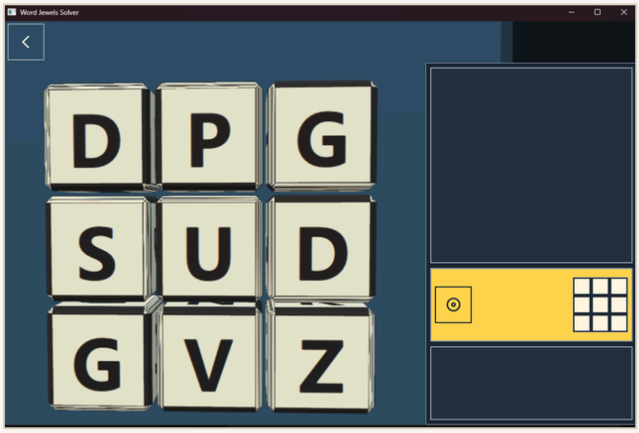


Exhibit 5. CountdownSolver and ICL Forge at the top; WordJewelsSolver home and Solo Play at the bottom. The gap is not “adult vs child”; it is “productised vs prototype”.

	COUNTDOWNSOLVER	ICL FORGE	WORDJEWELSSOLVER
Identity	C mark + Countdown SOLVER	Forge wordmark + product name	None; OS title bar only

	COUNTDOWNSOLVER	ICL FORGE	WORDJEWELSSOLVER
Type	IBM Plex Sans + Mono	Archivo + IBM Plex Mono	Andika named, not bundled; platform fallback
Accent	Teal	Signal red	Blue + amber + teal + five jewel colours
Shape	12 px cards	Zero radius	0 px child / 12 px parent
Labels	Every control	Every control + notes	Icon-only child controls
Empty states	Prompts what to do	Explains and guides	Blank cards; “Nothing to show yet.”
Productisation	Icon, version, packaging	Icon, first-run, docs	No icon, no font, no window state

BORROW

Wordmark, hierarchy, labels, empty states.

Countdown's page title and calm cards; Forge's editorial sections, mono labels, one accent and explanatory notes. Adapt them for a child: bigger, warmer, spoken.

DO NOT BORROW

Sidebar, density, adult tone.

A five-year-old does not need a navigation rail, a 1280 × 900 minimum or a 16 px settings row. Child-sized does not mean wordless.

THE DIRECTION

A jewel box on the table.

Warm paper, a deep tray, cream letter dice, an emerald accent and a gold reward. Flat editorial chrome around a tactile, physical play object. One product for the child and the parent, with the adult work behind the gate.

One obvious next step.

One primary action, labelled and reachable, on every screen.

Show, say, spell.

A picture, a word and a voice; never an icon alone for a child.

The board is the hero.

Frame it, light it, and settle it flat for reading and tracing.

A jewel for every win.

Visible, cumulative, satisfying; no score, no streak, no pressure.

Calm, not empty.

Every card has a title, a purpose and an empty state.

Grown-up jobs behind the gate.

Scanning, typing, settings and progress are adult workflows.

Paper
#F6F1E7

Ink
#1E2A30 · 13.05:1

Emerald
#166B5C · 5.67:1

Gold
#C98A1E · reward

Tray
#17323A

Die
#FFF7E3 · 13.73:1

Accent soft
#DDEDE8

Gold soft
#F7E8C6

DISPLAY ARCHIVO EXTRABOLD	**Word Jewels**
LEARNING ANDIKA	**cat · rain · star**
LABEL IBM PLEX MONO	3 × 3 · 4 WORDS · 12 JEWELS

Archivo and IBM Plex Mono are bundled with this brief under the SIL Open Font Licence; Andika is the target learning face (SIL OFL) and is shown here in a Verdana fallback until it is bundled in the app.

PROPOSED HOME — DIRECTION SKETCH, NOT FINAL UI

Word Jewels SOLVER — For grown-ups

A WORD-FINDING GAME FOR YOUNG READERS

Shake. Find. Collect.

S	O	S
S	A	N
N	S	L

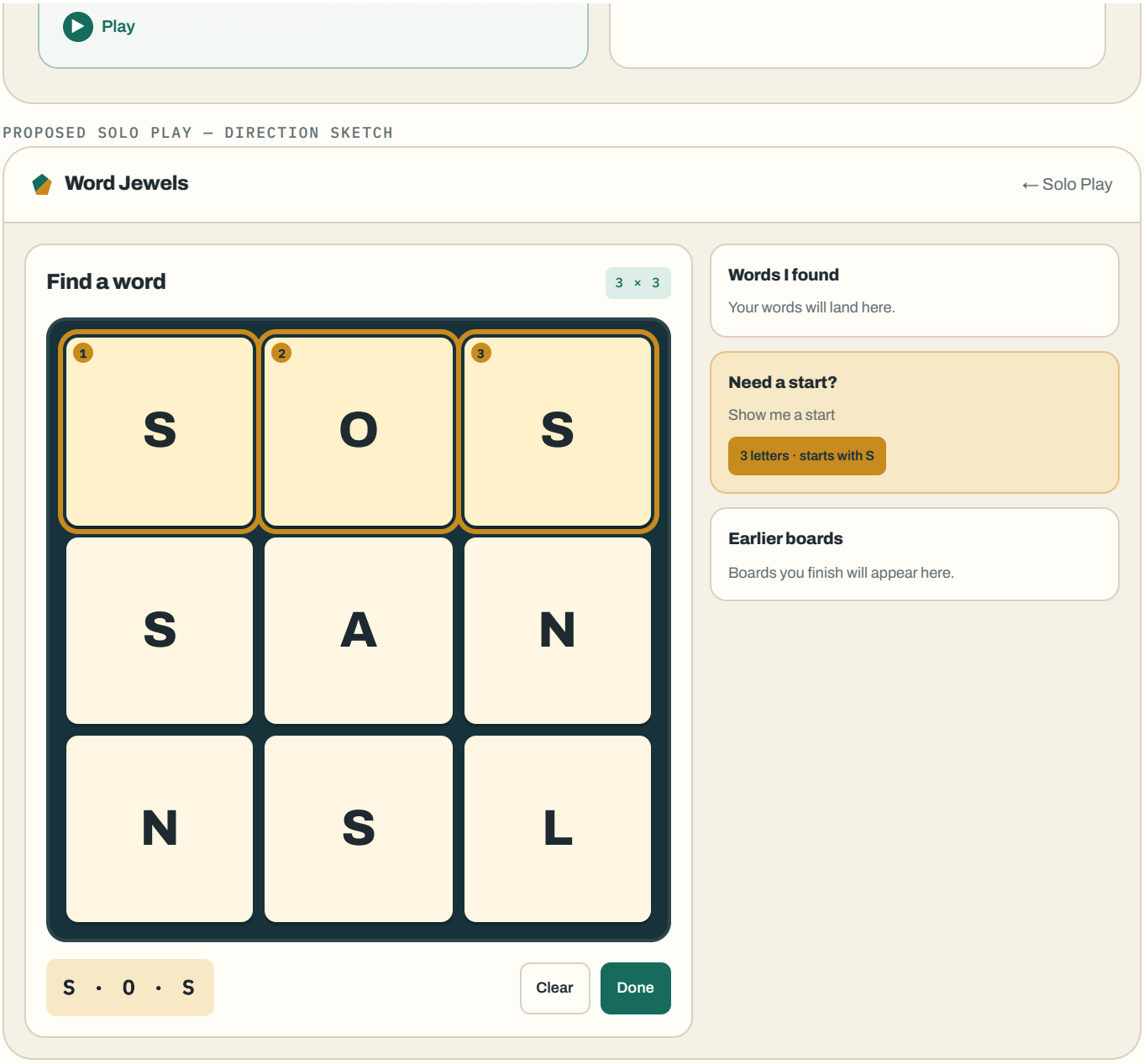
Play on my own

Shake the box. Find words. Collect jewels.

Play with a real board

A grown-up sets this up.

→ Continue



The sketch is a layout and hierarchy test, not a pixel spec. It shows the three moves that matter: a visible play label, a flat readable board, and a titled panel with an action zone.

THE PLAN

Four phases from prototype to product.

Ordered by user risk. Phase 0 removes the prototype signals and makes a child test possible; Phase 1 builds the design system and the loop; Phase 2 makes the family and parent workflows real; Phase 3 validates and polishes.

PHASE 0

Stop the prototype look

3–5 days

Wordmark, icon, home labels, panel titles and empty states, visible Done, path fix, clear summary, parent welcome, centred window, 2D fallback.

PHASE 1

One system + the child loop

2–3 weeks

Tokens, type, components, home, Solo box, flat board, titled panel, reward loop, gate and dashboard v1.

PHASE 2

Family + parent depth

3–4 weeks

Scan stepper, check screen, picker dialog, assist, parent onboarding, settings groups, content and recordings.

PHASE 3

Validate + release polish

2 weeks

5 children + 5 parents, accessibility, visual QA, performance, packaging, store assets.

> **Minimum lovable slice.** If the full overhaul is too much to start: identity + home labels + a visible box + a flat board + titled panel cards + a visible Done + a clear summary. That is enough to test whether a five-year-old can start, make a word, understand what happened, and stop.

THE NEXT DECISIONS

Five choices unlock the work.

The full audit has the screen-by-screen evidence, the code causes, the token tables, the validation script and the appendices. These are the decisions I need from you.

1. **Choose the role model.**

   One primary Play card with the adult workflows behind the gate (recommended), or two labelled cards with the family card gated. This changes the home.

2. **Approve the visual direction.**

   Jewel Box: warm paper, deep tray, emerald accent, gold reward, Archivo + Andika + Plex Mono, 12 px soft-modernist. Or a stricter Forge-like direction.

3. **Commit to the play surface.**

   Flat 2D for reading and tracing, 3D for the shake and drop. If 3D stays for play, it must become orthographic, seam-free and token-driven.

4. **Name the design owner.**

   A living style guide and one reviewer for every new screen. Otherwise the fourth vocabulary appears next sprint.

5. **Book the children.**

   Five 5–6 year olds and five parents after Phase 0/1. The biggest risk is not the solver; it is that the interface assumes a user who can already read it.

WHAT I WOULD DO FIRST

A one-week clarity pass.

Add a wordmark and app icon. Label the home and every child control. Put “Words I found” and an empty state on the panel. Move Done into a visible action zone. Fix the path. Add the clear summary. Centre the window. Then put it in front of one child and one parent before touching the 3D scene again.

Open the full audit →

Word Jewels Solver — Human Factors & Design Overhaul

19 October 2026 · Iain Chesworth Labs · Expert review, not a child study.

Full audit · Markdown source · Comparison image

Screenshots from the current Windows build; sibling screenshots from CountdownSolver and ICL Forge.